

• No class next Tuesday

• HW1 due October 2

• HW2 posted later today (due Oct 16)

Last class:

- Version control
- `git init`, `checkout`, `diff`, `status`, `add`, `commit`, `log`, `show`, `revert`

Today:

- using git across multiple machines

Now have 2 copies of our repo: local and remote. This adds 2 more steps to our workflow, when we synchronize our local copy with the remote one.

`git push` sends our local commits to the remote repository `git pull` brings commits from remote repository to our local copy

You can run these whenever you want, but it is most common to use them when you have local changes that you want to push.

```
scottmac@kaczmarz:~/CMSC6950/CMSC6950_Fall12025_examples$ git pull
Already up to date.
scottmac@kaczmarz:~/CMSC6950/CMSC6950_Fall12025_examples$ git add CMSC6950/schedule.txt
scottmac@kaczmarz:~/CMSC6950/CMSC6950_Fall12025_examples$ git commit -m"Add HW2 due date"
[main 221363c] Add HW2 due date
1 file changed, 1 insertion(+)
scottmac@kaczmarz:~/CMSC6950/CMSC6950_Fall12025_examples$ git push
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 16 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (4/4), 353 bytes | 353.00 KiB/s, done.
Total 4 (delta 0), reused 0 (delta 0), pack-reused 0
remote: resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:MacLachlanTeaching/CMSC6950_Fall12025_example.git
 48ff762..221363c  main -> main
```

Logic here is that you usually want to add your changes to an up-to-date copy of the repository (more on this later), so we pull any changes to the remote, add and commit as usual, then push your changes back to the repo.

IMPORTANT: `git push` and `git pull` only synchronize with changes that are committed. If you have local changes that you haven't committed, they won't get sent to the remote repo when you push.

Now that our repo is on github, we can access it from a different computer

```
math-scott02 (~) ~/CMSC6950/Fall12025> git clone git@github.com:MacLachlanTeaching/CMSC6950_Fall12025_example.git
Cloning into 'CMSC6950_Fall12025_example'...
remote: Enumerating objects: 38, done.
remote: Counting objects: 100% (38/38), done.
remote: Compressing objects: 100% (22/22), done.
remote: Total 38 (delta 3), reused 38 (delta 3), pack-reused 0 (from 0)
Receiving objects: 100% (38/38), done.
Resolving deltas: 100% (3/3), done.
math-scott02 (~) ~/CMSC6950/Fall12025> cd CMSC6950_Fall12025_example/
math-scott02 (~) ~/Fall12025/CMSC6950_Fall12025_example> ls
CMSC6950  data.log  myname.txt
```

syntax here is `git clone repository directory`. By default, `directory` is of same name as repository, but you can change this locally if you want.

Can now make changes in both copies of the remote repo. On my laptop:

```
math-scott02 (~) ~/Fall12025/CMSC6950_Fall12025_example> nano CMSC6950/schedule.txt
math-scott02 (~) ~/Fall12025/CMSC6950_Fall12025_example> git add CMSC6950/schedule.txt
```

```
math-scott02 (~/.Fall2025/CMS06950_Fall2025_example)> git commit -m"Add HW 3 to schedule"
[main 36ab0c1] Add HW 3 to schedule
1 file changed, 1 insertion(+)
math-scott02 (~/.Fall2025/CMS06950_Fall2025_example)> git push
```

```
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 10 threads
Compressing objects: 100% (3/3), done.
```

```

Writing objects: 100% (4/4), 364 bytes | 364.00 KiB/s, done.
Total 4 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:MacLachlanTeaching/CMSC6950_Fall2025_example.git
221363c..36ab0c1 main -> main

On workstation:

scott@mac$ git commit -m "CMSC6950/CMSC6950_Fall2025_example$ git pull
remote: Enumerating objects: 7, done.
remote: Counting objects: 100% (7/7), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 4 (delta 1), reused 4 (delta 1), pack-reused 0 (from 0)
Unpacking objects: 100% (4/4), 344 bytes | 114.00 KiB/s, done.
From github.com:MacLachlanTeaching/CMSC6950_Fall2025_example
221363c..36ab0c1 main
221363c..36ab0c1 main -> origin/main
Updating 221363c..36ab0c1
Fast-forward
 CMSC6950/schedule.txt | 1 +
 1 file changed, 1 insertion(+)

```

```
scottmac@kaczmarz:~/CHSC6950/CHSC6950_Fall2025_examples$ cat CHSC6950/schedule.txt
HW 2 out Sept 25
HW 1 due Oct 2
HW 3 out Oct 9
HW 1 due Oct 16

Can make edits in either place. On workstation:

scottmac@kaczmarz:~/CHSC6950/CHSC6950_Fall2025_examples$ nano CHSC6950/schedule.txt
HW 1 due Oct 2
scottmac@kaczmarz:~/CHSC6950/CHSC6950_Fall2025_examples$ git add CHSC6950/schedule.txt
scottmac@kaczmarz:~/CHSC6950/CHSC6950_Fall2025_examples$ git commit -m"Add HW3 due date:
> "
[main 1eb8d21] Add HW3 due date:
1 file changed, 1 insertion(+), 1 deletion(-)
scottmac@kaczmarz:~/CHSC6950/CHSC6950_Fall2025_examples$ git push
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 16 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (4/4), 365 bytes | 365.00 KiB/s, done.
Total 4 (delta 1), reused 0 (delta 0), pack-reused 0
Remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:MacLachlanTeaching/CHSC6950_Fall2025_example.git
36ab0c1..1eb8d21 main -> main
scottmac@kaczmarz:~/CHSC6950/CHSC6950_Fall2025_examples$ git log --oneline
1eb8d21 (HEAD -> main, origin/main) Add HW3 due date:
36ab0c1 Add HW 3 to schedule
221363c Add HW2 due date
48ff762 Adding data
902c4c0 Update gitignore
9c3b5fd Adding gitignore
61ec659 Revert "Add department"
eeff008 Adding HW2 to schedule
8ab3715 Add department
46dbd40 Adding schedule
a38ca54 Correcting university name
1413f06 Adding title
48ff00d Adding university
69db617 Add myname.txt

On laptop:

math-scott02 (~...Fall2025/CHSC6950_Fall2025_example)> git pull
remote: Enumerating objects: 7, done.
remote: Counting objects: 100% (7/7), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 4 (delta 1), reused 4 (delta 1), pack-reused 0 (from 0)
Unpacking objects: 100% (4/4), 345 bytes | 86.00 KiB/s, done.
From github.com:MacLachlanTeaching/CHSC6950_Fall2025_example
36ab0c1..1eb8d21 main -> origin/main
Updating 36ab0c1..1eb8d21
Fast-forward
 CHSC6950/schedule.txt | 2 +-
1 file changed, 1 insertion(+), 1 deletion(-)
math-scott02 (~...Fall2025/CHSC6950_Fall2025_example)> cat CHSC6950/schedule.txt
HW 2 out Sept 25
HW 1 due Oct 2
```

```

HW 3 out Oct 9
HW 2 due Oct 16
HW 3 due Oct 28
math-scott02 (~/.Fall2025/CMSC6950_Fall2025_example)> git log --oneline
1eb8621 (HEAD -- main, origin/main, origin/HEAD) Add HW3 due date
36a0b01 Add HW 3 to schedule
221363c Add HW2 due date
48ff762 Adding data
902c40d Update gitignore
9c39f5d Adding gitignore
61ec659 Revert "Add department"
eeff08d Adding HW2 to schedule
8ab3715 Add department
464db40 Adding schedule
a38ca54 Correcting university name
1413f06 Adding title
49fcb0b Adding university
69db617 Add myname.txt

```

When we work on the same repo from two different places, possible that our changes overlap. On laptop:

```

math-scott02 (~/.Fall2025/CMSC6950_Fall2025_example)> nano CMSC6950/schedule.txt
math-scott02 (~/.Fall2025/CMSC6950_Fall2025_example)> git add CMSC6950/schedule.txt
math-scott02 (~/.Fall2025/CMSC6950_Fall2025_example)> git commit -m"Adding HW4 to schedule"
[main 9a0a5e5] Adding HW4 to schedule
1 file changed, 1 insertion(+)
math-scott02 (~/.Fall2025/CMSC6950_Fall2025_example)> git push
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 10 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (4/4), 376 bytes | 376.00 KiB/s, done.
Total 4 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:MacLachlanfeaching/CMSC6950_Fall2025_example.git
1eb8621..9a0a5e5 main -> main

```

On workstation (at the same time):

```

scottmac@kaczmarz:~/CMSC6950/CMSC6950_Fall2025_example$ nano myname.txt
scottmac@kaczmarz:~/CMSC6950/CMSC6950_Fall2025_example$ git add myname.txt

```

```
scott@nagkaczmarz:~/CMS06950/CMS06950_Fall2025_examples$ git commit -m "Adding Department"
[main eb407a1] Adding Department
1 file changed, 1 insertion(+), 1 deletion(-)
scott@nagkaczmarz:~/CMS06950/CMS06950_Fall2025_examples$ git push
To github.com:MacLachlanTeaching/CMS06950_Fall2025_example.git
[rejected] main -> main (fetch first)
error: failed to push some refs to 'github.com:MacLachlanTeaching/CMS06950_Fall2025_example.git'
hint: Updates were rejected because the remote contains work that you do not
hint: have locally. This is usually caused by another repository pushing to
hint: the same ref. If you want to integrate the remote changes, use
hint: 'git pull' before pushing again.
hint: See the 'Note about fast-forward' in 'git push --help' for details
```

```

main:~$ git push --force
Push is rejected! GH "sees" that our local copy is out-of-sync with the remote and forces us to fix that before pushing.

On workstation:

scott@ckakmarz:~/CHSC6950/CHSC6950_Fall2025_examples$ git pull
remote: Enumerating objects: 7, done.
remote: Counting objects: 100% (7/7), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 4 (delta 1), reused 4 (delta 1), pack-reused 0 (from 0)
Unpacking objects: 100% (4/4), 356 bytes | 118.00 KiB/s, done.
From github.com:MacLachlanTeaching/CHSC6950_Fall2025_example
  1eb8021..59b85e5  main                origin/main
Merge made by the 'ort' strategy.
 CHSC6950/schedule.txt | 1 +
 1 file changed, 1 insertion(+)
scott@ckakmarz:~/CHSC6950/CHSC6950_Fall2025_examples$ git status

```

```

On branch main
Your branch is ahead of 'origin/main' by 2 commits.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean
scottmac@kaccharr:~/CSC60950/CSC60950_Fall2025_examples$ git push
Enumerating objects: 9, done.
Counting objects: 100% (8/8), done.
Delta compression using up to 16 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (5/5), 651 bytes | 325.00 KiB/s, done.
Total 5 (delta 2), reused 0 (delta 0), packed 3 (delta 0)

```

```

Remote: Resolving dates: 100% (2/2), completed with 2 local objects.
To github.com:MacLachlanTeaching/CHSC6950_Fall2025_example.git
9a8a5e5..22592d2 main -> main
scottmac@kczarrz:~/CHSC6950/CHSC6950_Fall2025_examples git log --oneline
225b261 (HEAD -> main, origin/main) Merge branch 'main' of github.com:MacLachlanTeaching/CHSC6950_Fall2025_example
eb9a7f1 Adding Department
9a8a5e5 Adding HW4 to schedule
1eb8d21 Add HW3 due date:
36a8bc1 Add HW 3 to schedule
221363c Add HW2 due date
4bf7f62 Adding data
0876c0b Update distrover

```

```
9c39f5d Adding gitignore
61ec659 Revert "Add department"
ceef008 Adding hw2 to schedule
8ab3715 Add department
464bd40 Adding schedule
a38ca54 correcting university name
1413f06 Adding title
48fdbd0 Adding university
69db617 Add myname.txt
```

*Merging is how github manages changes that aren't pulled before committing. The above is the easiest case. We made changes to 2 different files, so no problems merging.

Let's try something more dangerous.

On laptop:

```
math-scott02 [....Fall2025/CHSC6950_fall2025_example]> git pull
remote: Enumerating objects: 9, done.
remote: Counting objects: 100% (8/8), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 5 (delta 2), reused 5 (delta 2), pack-reused 0 (from 0)
```

```
Unpacking objects: 100% (5/5), 631 bytes | 126.00 KiB/s, done.
From github.com:MacLachlanTeaching/CMS6950_Fall2025_example
 90a565..22502a1 main -> origin/main
Updating 90a565..22502a1
Fast-forward
 nymath.txt | 1 +
 1 file changed, 1 insertion(+)
math-scott02 (~/.ssh/CMS6950_Fall2025_example): nano CMS6950/schedule.txt
math-scott02 (~/.ssh/CMS6950_Fall2025_example): git add CMS6950/schedule.txt
math-scott02 (~/.ssh/CMS6950_Fall2025_example): git commit -m"Adding project proposals"
 [main e01055] Adding project proposals
 [main changed] 1 insertion(+)

```

```
math-scott02 [---Fall2025/CMSC6950_Fall2025_example]# git push
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 10 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (4/4), 395 bytes | 395.00 KiB/s, done.
Total 4 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:MacIsachan/teaching/CMSC6950_Fall2025_example.git
225b2a1..e011055 main -> main
```

```

UNION workstation:
scott@aczkazmarz:~/CMSC6950/CMSC6950_Fall2025_examples nano CMSC6950/schedule.txt
scott@aczkazmarz:~/CMSC6950/CMSC6950_Fall2025_examples git add CMSC6950/schedule.txt
scott@aczkazmarz:~/CMSC6950/CMSC6950_Fall2025_examples git commit -m"Adding H44 due date"
[main 80462a2] Adding H44 due date
1 file changed, 1 insertion(+)
scott@aczkazmarz:~/CMSC6950/CMSC6950_Fall2025_examples git push
To github.com:MacLachlanTeaching/CMSC6950_Fall2025_example.git
 ! [rejected]        main -> main (fetch first)
error: failed to push some refs to 'github.com:MacLachlanTeaching/CMSC6950_Fall2025_example.git'
tips: updates were rejected because the remote contains work that you do not

```

```
hint: have locally. This is usually caused by another repository pushing to
hint: the same ref. If you want to integrate the remote changes, use
hint: 'git pull' before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.

Again rejected

scottmac@kaczmarz:~/CMS66950/CMS66950_Fall2025_examples git pull
remote: Enumerating objects: 7, done.
remote: Counting objects: 100% (7/7), done.
remote: Compressing objects: 100% (7/7), done.
```

```
remote: Total 4 (delta 1), reused 4 (delta 1), pack-reused 0 (from 0)
Unpacking objects: 100% (4/4), 375 bytes | 125.00 KiB/s, done.
From github.com:MacLachlanTeaching/CMSC6950_Fall2025_example
225b2a1..c01105f main -> origin/main
Auto-merging CMSC6950/schedule.txt
Merge made by the 'ort' strategy.
CMSC6950/schedule.txt | 1 +
1 file changed, 1 insertion(+)
scott@maczkamzr:~/CMSC6950/CMSC6950_Fall2025_examples$ git push
Enumerating objects: 14, done.
Counting objects: 100% (14/14), done.
```

```
Delta compression using up to 16 threads
Compressing objects: 100% (8/8), done
Writing objects: 100% (8/8), 729 bytes | 243.00 KiB/s, done.
Total 8 (delta 4), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (4/4), completed with 2 local objects.
To github.com:MacLachlanTeaching/CMS6950_Fall1825_example.git
c010555..99a20ee main -> main
scott@scmcarz:~/CMS6950/CMS6950_Fall1825_examples$ git log --oneline -N 5
fatal: ambiguous argument '5': unknown revision or path not in the working tree.
Use '-' to separate paths from revisions, like this:
'git  [...] -- [...]'
scott@scmcarz:~/CMS6950/CMS6950_Fall1825_examples$ git log --oneline -N 5
```

```
09b26ee HEAD => main, origin/main Merge branch 'main' of github.com:MacLachlanTeaching/CMS6950_Fall2025_example
09b26ee Adding HW due Oct 21
e0105d5 Adding project proposals
225b241 Merge branch 'main' of github.com:MacLachlanTeaching/CMS6950_Fall2025_example
e0b87d1 Adding Department
sc02b6ac@cmzarr:~/CMS6950/CMS6950_Fall2025_examples$ cat CMS6950/schedule.txt
HW 2 out Sept 25
HW 1 due Oct 2
HW 3 out Oct 9
HW 2 due Oct 16
Project proposals due Oct 21
HW 4 out Oct 23
```

```
HW 3 due Oct 28
HW 4 due November 6
```

At the end of this, both changes are there since they didn't "conflict". We made 2 changes in different parts of the same file, so git was still able to "auto-merge" them.

Go back to out-of-date copy on laptop"

```
math-scott02 (~/.Fall2025/CMS6950_Fall2025_example)~ nano CMS6950/schedule.txt
math-scott02 (~/.Fall2025/CMS6950_Fall2025_example)~ git add CMS6950/schedule.txt
math-scott02 (~/.Fall2025/CMS6950_Fall2025_example)~ git commit -m"Adding HWs to schedule"
[main a3cf8bce] Adding HWs to schedule
```

```
1 file changed, 1 insertion(+)  
math-science@102 (~) [~/Falls2025/MSC6590_Fall2025_example] > git push  
to github.com:MacLachlanTeaching/MSC6590_Fall2025_example.git  
[rejected]          main -> main (fetch first)  
error: failed to push some refs to 'github.com:MacLachlanTeaching/MSC6590_Fall2025_example.git'  
hint: Updates were rejected because the remote contains work that you do  
hint: have locally. This is usually caused by another repository pushing  
hint: to the same ref. You may want to first integrate the remote changes  
hint: (e.g., 'git pull ...') before pushing again.  
hint: See the 'Note about fast-forwards' in 'git push --help' for details.  
math-science@102 (~) [~/Falls2025/MSC6590_Fall2025_example] > git pull  
remote: Enumerating objects: 14, done.
```

```

remote: Counting objects: 100% (14/14), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 8 (delta 4), reused 8 (delta 4), pack-reused 0 (from 0)
Unpacking objects: 100% (8/8), 789 bytes | 138.0 KiB/s, done.
from github.com:MacLachlanTeaching/CSC6050_Fall1625_example
011055..99b26ee main -> origin/main
Auto-merging CSC6050/schedule.txt
CONFLICT (content): Merge conflict in CSC6050/schedule.txt
Automatic merge failed; fix conflicts and then commit the result.

```

This leads to a merge conflict. Instructions tell us that **we need to "fix conflicts and then commit the result"**

```

W 2 out Sept 25
HW 1 due Oct 2
HW 3 out Oct 9
HW 2 due Oct 16
Project proposals due Oct 21
HW 4 out Oct 23
HW 3 due Oct 28
===== HW40
HW 5 out Nov 4
=====
HW 6 out - November 6

```

```
git diff --no-pager
diff --git a/math-scott02/Fall2025/CHSC6950_Fall2025_example.py b/math-scott02/Fall2025/CHSC6950_Fall2025_example.py
index 7d178720f0171b25914db078617ada1..09b26ecb79d178720f0171b25914db078617ada1
```

Have choices here:

1. Choose one change (local, marked HEAD, or remote, marked by hash)
2. Take both changes (and just delete git markings)
3. Do something else. Just manually edit file until it is in a state that you're willing to commit.

```
math-scott02 (~---Fall2025/CHSC6950_Fall2025_example)» git status
```

```

Your branch and 'origin/main' have diverged,
and have 1 and 2 different commits each, respectively.
(use "git pull" to merge the remote branch into yours)

You have unmerged paths.
(fix conflicts and run "git commit")
(use "git merge --abort" to abort the merge)

Unmerged paths:
  (use "git add <file>..." to mark resolution)
    both modified:   CMS6950/schedule.txt

```

no changes added to commit (use "git add" and/or "git commit -a")

Note the instructions: can abort the merge or use "git add" to mark resolution

On laptop:

```
math-scott02 (~) Fall2025/CMSC6950_Fall2025_example> git add CMSC6950/schedule.txt
math-scott02 (~) Fall2025/CMSC6950_Fall2025_example> git commit -m"fixing merge, keeping both changes"
[main b5afaa6] Fixing merge, keeping both changes
1 file changed, 1 insertion(+), 1 deletion(-)
math-scott02 (~) Fall2025/CMSC6950_Fall2025_example> git push
```

```
Enumerating objects: 100% (1/1), done.
Counting objects: 100% (14/14), done.
Delta compression using up to 10 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (9/9), 711 bytes | 711.00 KiB/s, done.
Total 8 (delta 4), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (4/4), completed with 2 local objects.
To github.com:MarcelJanLaTeaching/OSCS6950_Final2025_example.git
09b26ec..b6afa66 main -> main
math-schott02 [~/.../Final2025/OSCS6950_Final2025_example]: git log --oneline -n 5
b6afa66 (HEAD -- main, origin/main, origin/HEAD) Fixing merge, keeping both changes
43cf06e Adding HWS to schedule
```

```
09526ce Merge branch 'main' of github.com:MacLachlanTeaching/CMSC6950_Fall12025_example
000624a Adding HW4 due date
e011055 Adding project proposals

What should I have done? Pulled before adding/committing.

On workstation:

scottmac@kaczmarz:~/CMSC6950/CMSC6950_Fall12025_examples$ nano CMSC6950/schedule.txt
scottmac@kaczmarz:~/CMSC6950/CMSC6950_Fall12025_examples$ git pull
remote: Enumerating objects: 14, done.
remote: Counting objects: 100% (14/14), done.
```

```

remote: Compressing objects: 100% (2/2), done.
remote: Total 8 (delta 4), reused 8 (delta 4), pack-reused 0 (from 0)
Unpacking objects: 100% (8/8), 691 bytes | 130.00 KiB/s, done.
From github.com:MacLachlanTeaching/CMSC6950_Fall2025_example
 9b026ee..db4faa6 main      -> origin/main
Updating 9b026ee..db4faa6
error: Your local changes to the following files would be overwritten by merge:
       CMSC6950/schedule.txt
Please commit your changes or stash them before you merge.
Aborting

```

Local changes would be overwritten by merge. Commit or stash them first.

We know if we commit the changes, we will have to resolve a merge conflict. So try the other way

```
scottmac@kaczmarz:~/CMSC6950/CMSC6950_Fall1025_examples$ git stash
Saved working directory and index state WIP on main: 0b926e6 Merge branch 'main' of github.com:MacLachlanTeaching/CMSC6950_Fall1025_examples
scottmac@kaczmarz:~/CMSC6950/CMSC6950_Fall1025_examples$ git status
On branch main
Your branch is behind 'origin/main' by 2 commits, and can be fast-forwarded.
(use "git pull" to update your local branch)
```

```
nothing to commit, working tree clean
scott@kaczmarz:~/CSC6950/CSC6950_Fall2025_examples$ git stash list
stash@0: WIP on main: 09b26ee Merge branch 'main' of github.com:MatthachlanTeaching/CSC6950_Fall2025_examples

Changes are "stashed" and my repository is ready to merge from the remote. Now pull and "apply changes"

On workstation:

scott@kaczmarz:~/CSC6950/CSC6950_Fall2025_examples$ git pull
Updating 09b26ee..b6afa86
Fast-forward
```

```

C:\CMSC6950\schedule.txt | 1
1 file changed, 1 insertion(+)
scott@ac-kaczmarcz:~/CMSC6950/CMSC6950_Fall2025_examples$ git stash apply
Auto-merging CMSC6950/schedule.txt
On branch main
Your branch is up to date with 'origin/main'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
    modified:   CMSC6950/schedule.txt

```

```
no changes added to commit (use "git add" and/or "git commit -a")
scott@bkacmarz:~/CHSC6950/CHSC6950_Fall2025_examples$ cat CHSC6950/schedule.txt
HW 2 due Sept 25
HW 1 due Oct 2
HW 3 due Oct 9
HW 2 due Oct 16
Project proposals due Oct 21
HW 4 due Oct 23
HW 3 due Oct 28
HW 5 out Nov 4
HW 4 due November 6
HW 6 out November 13
```

```
scott@kacmarzr:~/CMSG950/CMSC950_Fall12025_examples$ git add CMSC950/schedule.txt
scott@kacmarzr:~/CMSG950/CMSC950_Fall12025_examples$ git commit -m "Adding Hm6 to schedule"
[main 3f207b6] Adding Hm6 to schedule
1 file changed, 1 insertion(+)
scott@kacmarzr:~/CMSG950/CMSC950_Fall12025_examples$ git push
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 16 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (4/4), 368 bytes | 366.00 KiB/s, done.
Total 4 (delta 2), reused 0 (delta 0), pack-reused 0
remote: Compressing objects: 100% (3/3), done.
remote: Total 4 (delta 2), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (2/2), done.
scott@kacmarzr:~/CMSG950/CMSC950_Fall12025_examples$
```

```

$ git checkout -b 12/27/2025 main
$ git push
To github.com:MacLachlanTeaching/OSC6950_Fall2025_example.git
 b6af8a6..3f2d7b6 main -> main
scottcackmaczmarz@OSC6950/OSC6950_Fall2025_example git log --one-line -n 6
3f2d7b6 (HEAD -> main, origin/main) Adding HW5 to schedule
b6af8a6 Fixing merge, keeping both changes
a3cfb0e Adding HW5 to schedule
09b02ee Merge branch 'main' of github.com:MacLachlanTeaching/OSC6950_Fall2025_example
080462a Adding HW4 due date
e018055 Adding project proposals

```

When we want to make bigger changes (or commit to larger projects), we can use "branches" to manage merging in a bit of a cleaner way. Will discuss this in a few classes.

One of our main motivations for using `git` is that it provides tools for **reproducible research**.

We often think of scientific manuscripts as the "output" of the scientific process. That is, the knowledge gained by "doing science" (the scholarship) is the manuscript itself.

This is wrong

"An article about computational science in a scientific publication is not the scholarship itself, it is merely advertising of the scholarship. The actual scholarship is the complete software development environment and the complete set of instructions which generated the figures."

The science that you do when running computational experiments is concentrated in the computer code itself.

To do good science, we must write good code.

What is good code?

Good code is:

- readable
- runnable
- correct
- documented

- open (everyone can examine it)

What makes code readable?

- A knowledgeable user should be able to easily understand what it does
- Our memories work best with small chunks of data at once, so write code in this style too. Break large computational tasks into small ones. Each function does just one thing, acting on just data needed for that task.
- Variable names and coding style should be consistent. Variable names should be meaningful.
- Do not repeat code. Use functions do essentially the same thing, combine them. Don't make the reader guess at what's different. Also helps when fixing bugs. If code exists only in one place, then it can be fixed in one place.

- Do not repeat data. Physical constants, important parameters, data file names, etc., should be set **once** then passed to relevant functions when needed. Should not be redefined in many places. Don't re-read search your code to find where you define a critical parameter.
- Don't reinvent the wheel. Many high-quality implementations of common utility functions (searching/sorting, data structures) and mathematical operations (numerical approximations, optimization, linear algebra) exist. It is more efficient for you to use these libraries than it is for you to reimplement their functionality in your own code.